Math Inquirers

# Extension Workbook

## with answers

**Level 3**

Self-study | Ages 8-10




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 3**
Self-study practice with answers.


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## Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

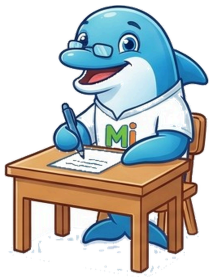

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# Welcome!




This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

## How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

## For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

## The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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| 3 | **Times table treasure**<br>Not in this proof edition | – |
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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | Not in this proof edition | | | | ☐ |
| 2 | Cryptarithm crackers | Not in this proof edition | | | | ☐ |
| 3 | Times table treasure | Not in this proof edition | | | | ☐ |
| 4 | Model makers (bar models) | Not in this proof edition | | | | ☐ |
| 5 | Remainder riddles | Not in this proof edition | | | | ☐ |
| 6 | Fraction walls | Not in this proof edition | | | | ☐ |
| 7 | Area architects | Not in this proof edition | | | | ☐ |
| 8 | Angle and line detectives | Not in this proof edition | | | | ☐ |
| 9 | Measure and money masters | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 10 | Puzzle day | Not in this proof edition | | | | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

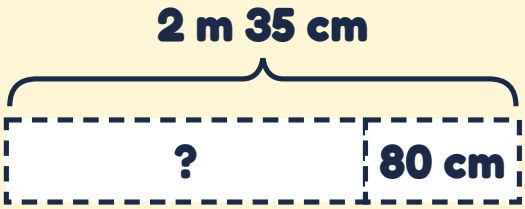
# Learn it: Length, mass, volume, money and time

**My goal.** I can change units, solve money problems and find how long things last.

Use one unit before you add or take away. 1 m = 100 cm, 1 kg = 1000 g, 1 ℓ = 1000 ml and $1 = 100¢.

## Worked example 1

**A ribbon is 2 m 35 cm long. Mei cuts off 80 cm. How long is the ribbon now?**

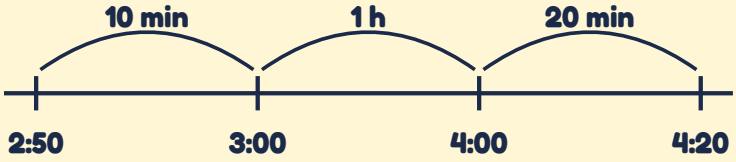



1. Change to cm: 2 m 35 cm = 235 cm.
2. Take away: 235 − 80 = 155 cm = 1 m 55 cm.

**Answer: The ribbon is 1 m 55 cm long.**

## Worked example 2

**A film starts at 2:50 pm and ends at 4:20 pm. How long is the film?**




1. Jump to the next o'clock, then to 4:00, then to 4:20.
2. Add the jumps: 10 min + 1 h + 20 min = 1 h 30 min.

**Answer: The film is 1 h 30 min long.**

## Tips

1. Write both amounts in the small unit first.
2. On a time line, jump to the next o'clock.

## Your turn

Answer at the back of the book.

**Try** A jug holds 2 ℓ 350 ml. Ali pours out 800 ml. The jug now holds ☐ ℓ ☐ ml.

## Start here

Do these first. Take your time.

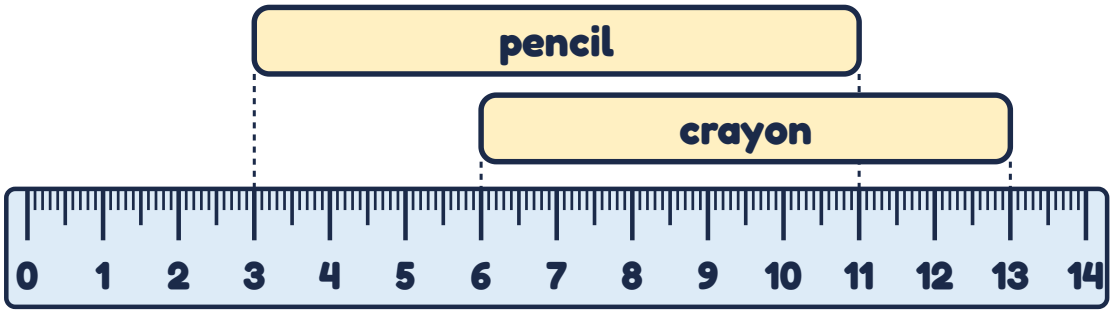
**1** **Change the units.**

(a) 3 m = ☐ cm (b) 4 kg 50 g = ☐ g

(c) 5 ℓ 300 ml = ☐ ml (d) 2750 g = ☐ kg ☐ g

(e) 365 cm = ☐ m ☐ cm

**2** **The ruler is shown at real size. Read the length of each object.**




(a) The pencil is ☐ cm long.

(b) The crayon is ☐ cm long.

(c) The pencil is ☐ cm longer than the crayon.

**3** **The parcels weigh: A 1 kg 80 g, B 1900 g, C 1 kg 8 g, D 1 kg 90 g. Write the letters in order. Begin with the lightest.**

☐ ☐ ☐ ☐

**4** **A jug has 2 ℓ 500 ml of juice. Ali pours it into cups of 400 ml.**

(a) He fills ☐ cups.

(b) ☐ ml of juice are left over.

## Start here

Keep going. Check your work.

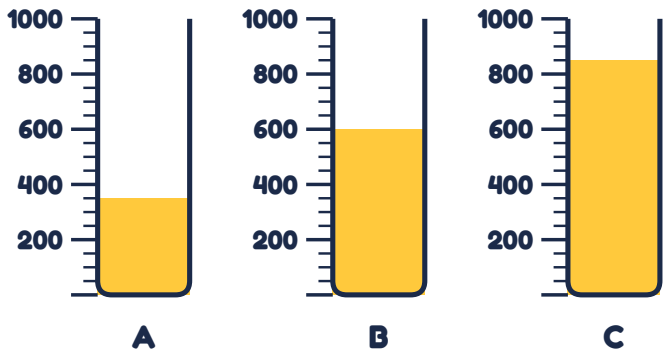
**5** **Read the water level in each jug.**

(a) A holds ☐ ml

(b) B holds ☐ ml

(c) C holds ☐ ml

(d) C has ☐ ml more than A




**6** **A notebook costs $2.65 and a ruler costs $1.40. Mei pays with a $5 note.**

(a) They cost $ ☐ in all.

(b) Her change is $ ☐ .

**7** **Siti has $20 and buys the first 3 things on the list.**

| Book | Puzzle | Bottle | Stickers |
|---|---|---|---|
| $6.20 | $4.85 | $3.40 | $1.75 |

(a) They cost $ ☐ in all.

(b) Her change is $ ☐ .

(c) Can she buy all 4 things on the list? Yes No

**8** **A box with 6 books weighs 2 kg 250 g. The empty box weighs 450 g. All the books have the same mass. One book weighs ☐ g.**

Working: _____

## Start here

Nearly there. You can do it.

**9** **Write the time in the other kind of time.**

(a) 6:15 am = ☐ (b) 1:40 pm = ☐ (c) 9:05 pm = ☐

(d) 16:50 = ☐ pm (e) 20:25 = ☐ pm

**10** **How long does each take? Use a time line to help.**

(a) Aisha's class: 5:25 pm to 6:40 pm. It lasts ☐ h ☐ min.

(b) Ben's class starts at 3:40 pm and lasts 55 minutes. It ends at ☐.

(c) A show ends at 8:15 pm. It is 1 h 45 min long. It starts at ☐.

Time line: _____

**11** **Write the time shown on each clock.**




(a) A: ☐ (b) B: ☐ (c) C: ☐

**12** **A stamp costs 65¢. Kumar buys 8 stamps.**

(a) The stamps cost $ ☐.

(b) He pays with a $10 note. His change is $ ☐.

## Think harder

Hints are upside down. Try first!

**1** **The buses leave the interchange every 20 minutes, starting at 06:50. The ride takes 25 minutes. Mei must reach school by 08:15. When does the latest bus she can take leave?** ☐

Bus times: 06:50, _____

_____

_____

**Hint.** Find the latest leaving time first: 08:15 minus 25 minutes.

**2** **A night bus leaves at 22:50 and arrives at 02:10 the next day. How long is the trip?** ☐ **h** ☐ **min**

Time line: _____

_____

_____

**Hint.** Jump to midnight first. Then jump to 02:10.

**3** **2 notebooks and 3 pens cost $9. A notebook costs $1.50 more than a pen.**

**(a)** A pen costs $ ☐.

**(b)** A notebook costs $ ☐.

Working: _____

_____

_____

**Hint.** Take away the extra $1.50 for each notebook first.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Beyond Level 3: A water puzzle.**
**You have a 3 ℓ jug, a 5 ℓ jug and a tap. The jugs have no marks. You want exactly 4 ℓ.**

(a) Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

(b) Empty the 3 ℓ jug. Pour the 2 ℓ into it. Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

**Hint.** Follow each step. The jug is full when it holds 3 ℓ.

**2** **Mei's watch is 8 minutes slow.**

(a) Her watch shows 4:20 pm. The real time is ☐.

(b) A film starts at 4:45 pm. When it starts, her watch shows ☐.

**Hint.** A slow watch shows an earlier time than the real time.

**3** **Write a shopping story that uses $6.40 and $2.15. The answer to your story must be $4.25.**

My story: _____

_____

My answer: _____

**Hint.** Which sum or take-away of $6.40 and $2.15 gives $4.25?

**4** **A film lasts 1 h 55 min. It ends at 00:20, just after midnight. At what time did it start? Write it in 24-hour time.** ☐

Time line: _____

**Hint.** Go back to midnight first. Then go back the rest.

## ✔ Check yourself

No hints. Try each one on your own.

**1** 2 m 40 cm = ☐ cm

Working: _____

**2** A bottle holds 1 ℓ 200 ml. A child drinks 700 ml. The bottle then holds ☐ ml.

Working: _____

**3** A film starts at 6:45 pm and ends at 8:10 pm. It lasts ☐ h ☐ min.

Working: _____

**4** Kumar buys a snack for $2.35. He pays with a $5 note. His change is $ ☐ .

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Cryptarithm crackers | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Times table treasure | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Model makers (bar models) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Remainder riddles | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Fraction walls | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Area architects | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Angle and line detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Measure and money masters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Puzzle day | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 3.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 9 · Measure and money masters

**Your turn**
page 6

**Answer: 1 ℓ 550 ml**
**Why:** 2 ℓ 350 ml = 2350 ml. Then 2350 - 800 = 1550 ml = 1 ℓ 550 ml. Check: 1550 + 800 = 2350.

## START HERE

**1**
page 7

**Answer: (a) 300 (b) 4050 (c) 5300 (d) 2 kg 750 g (e) 3 m 65 cm**
**Why:** (a) 3 × 100 = 300. (b) 4 kg = 4000 g, and 4000 + 50 = 4050. (c) 5 ℓ = 5000 ml, and 5000 + 300 = 5300. (d) 2000 g is 2 kg, and 750 g is left. (e) 300 cm is 3 m, and 65 cm is left.

**2**
page 7

**Answer: (a) 8 (b) 7 (c) 1**
**Why:** The pencil starts at 3 and ends at 11: 11 - 3 = 8 cm. The crayon starts at 6 and ends at 13: 13 - 6 = 7 cm. The pencil is 8 - 7 = 1 cm longer.

**3**
page 7

**Answer: C, A, D, B**
**Why:** Write each in grams: A is 1080 g, B is 1900 g, C is 1008 g and D is 1090 g. In order: 1008, 1080, 1090, 1900. That is C, A, D, B.

**4**
page 7

**Answer: (a) 6 (b) 100**
**Why:** 2 ℓ 500 ml = 2500 ml. 2500 ÷ 400 = 6 remainder 100. He fills 6 cups, and 100 ml are left over.

**5**
page 8

**Answer: (a) 350 (b) 600 (c) 850 (d) 500**
**Why:** Each small mark is 50 ml. A is 1 mark above 300: 350 ml. B is at 600 ml. C is 1 mark above 800: 850 ml. 850 - 350 = 500 ml.

**6**
page 8

**Answer: (a) 4.05 (b) 0.95**
**Why:** (a) 2.65 + 1.40 = 4.05. (b) 5.00 - 4.05 = 0.95. Check: 4.05 + 0.95 = 5.00.

**7**
page 8

**Answer: (a) 14.45 (b) 5.55 (c) Yes**
**Why:** (a) 6.20 + 4.85 = 11.05, and 11.05 + 3.40 = 14.45. (b) 20.00 - 14.45 = 5.55. (c) All 4 things cost 14.45 + 1.75 = 16.20, which is less than $20. So she can.

**8**
page 8

**Answer: 300**
**Why:** 2 kg 250 g = 2250 g. The books weigh 2250 - 450 = 1800 g. One book: 1800 ÷ 6 = 300 g.

**9**
page 9

**Answer: (a) 06:15 (b) 13:40 (c) 21:05 (d) 4:50 (e) 8:25**
**Why:** After 12 noon, add 12 to the hour for 24-hour time. (b) 1 + 12 = 13. (c) 9 + 12 = 21. For (d) and (e), take away 12: 16 - 12 = 4 and 20 - 12 = 8.

**10**
page 9

**Answer: (a) 1 h 15 min (b) 4:35 pm (c) 6:30 pm**
**Why:** (a) 5:25 to 6:00 is 35 min, then 6:00 to 6:40 is 40 min. 35 + 40 = 75 min = 1 h 15 min. (b) 3:40 + 20 min = 4:00, then 35 more min = 4:35 pm. (c) 8:15 pm - 1 h = 7:15 pm, then 45 min back = 6:30 pm.

**11** page 9

**Answer: (a) 4:10 (b) 7:45 (c) 11:25**
**Why:** The short hand shows the hour and the long hand shows the minutes. A: the long hand points at 2, which is 10 minutes, and the short hand is just past 4. B: the long hand points at 9, which is 45 minutes. C: the long hand points at 5, which is 25 minutes.

**12** page 9

**Answer: (a) 5.20 (b) 4.80**
**Why:** (a) 8 × 65¢ = 520¢ = $5.20. (b) 10.00 - 5.20 = 4.80.

## THINK HARDER

**1** page 10

**Answer: 07:50**
**Why:** Mei must leave by 08:15 - 25 min = 07:50. The buses leave at 06:50, 07:10, 07:30, 07:50, 08:10. The latest bus that gets her to school in time is the 07:50 bus.

**2** page 10

**Answer: 3 h 20 min**
**Why:** From 22:50 to 24:00 (midnight) is 1 h 10 min. From midnight to 02:10 is 2 h 10 min. 1 h 10 min + 2 h 10 min = 3 h 20 min.

**3** page 10

**Answer: (a) 1.20 (b) 2.70**
**Why:** The 2 notebooks cost $3 more than 2 pens in all (2 × $1.50). Take that away: $9 - $3 = $6 for 5 pens. One pen costs $6 ÷ 5 = $1.20. A notebook costs $1.20 + $1.50 = $2.70. Check: 2 × 2.70 + 3 × 1.20 = 5.40 + 3.60 = 9.00.

## GO FURTHER

**1** page 11

**Answer: (a) 2 (b) 4**
**Why:** (a) The 3 ℓ jug takes 3 ℓ, so 5 - 3 = 2 ℓ are left in the 5 ℓ jug. (b) The 3 ℓ jug holds 2 ℓ and has room for 1 ℓ more. The full 5 ℓ jug gives 1 ℓ, so 5 - 1 = 4 ℓ are left. That is exactly 4 ℓ.

**2** page 11

**Answer: (a) 4:28 pm (b) 4:37 pm**
**Why:** (a) A slow watch is behind: the real time is 8 minutes later: 4:20 pm + 8 min = 4:28 pm. (b) At the real time 4:45 pm her watch shows 8 minutes earlier: 4:37 pm.

**3** page 11

**Answer: Answers vary. Example: A book costs $6.40 and a pen costs $2.15. How much more does the book cost than the pen? The answer is $4.25.**
**Why:** Check that your story uses both amounts and gives $4.25. The only way is the difference: 6.40 - 2.15 = 4.25. A sum of the two would be $8.55.

**4** page 11

**Answer: 22:25**
**Why:** From 00:20 go back 20 min to midnight (24:00). 1 h 55 min - 20 min = 1 h 35 min is left. 24:00 - 1 h 35 min = 22:25. Check: 22:25 + 1 h 55 min = 00:20.

## CHECK YOURSELF

**1** page 12

**Answer: 240**
**Why:** 2 m = 200 cm, and 200 + 40 = 240 cm.

**2**
page 12

**Answer: 500**
**Why:** 1 ℓ 200 ml = 1200 ml. Then 1200 - 700 = 500 ml.

**3**
page 12

**Answer: 1 h 25 min**
**Why:** 6:45 to 7:00 is 15 min. 7:00 to 8:00 is 1 h. 8:00 to 8:10 is 10 min. In all: 1 h 25 min.

**4**
page 12

**Answer: 2.65**
**Why:** 5.00 - 2.35 = 2.65. Check: 2.35 + 2.65 = 5.00.